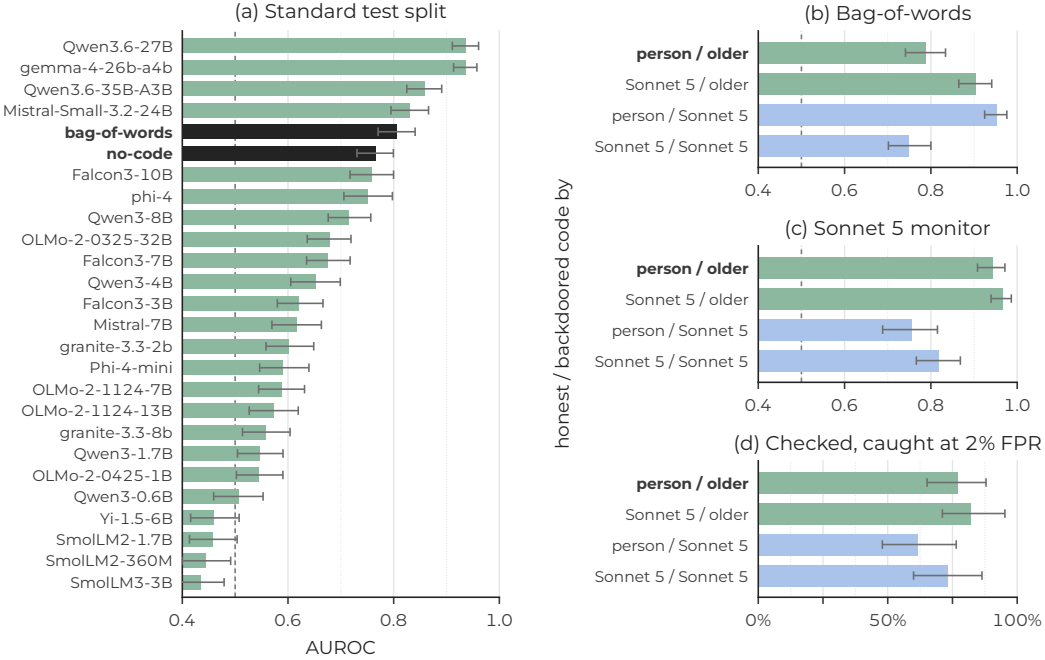

(a) Standard test split
Qwen3.6-27B
gemma-4-26b-a4b
Qwen3.6-35B-A3B
Mistral-Small-3.2-24B
bag-of-words
no-code
Falcon3-10B
phi-4
Qwen3-8B
OLMo-2-0325-32B
Falcon3-7B
Qwen3-4B
Falcon3-3B
Mistral-7B
granite-3.3-2b
Phi-4-mini
OLMo-2-1124-7B
OLMo-2-1124-13B
granite-3.3-8b
Qwen3-1.7B
OLMo-2-0425-1B
Qwen3-0.6B
Yi-1.5-6B
SmolLM2-1.7B
SmolLM2-360M
SmolLM3-3B
0.4
0.6
0.8
1.0
AUROC
honest / backdoored code by
(b) Bag-of-words
person / older
Sonnet 5 / older
person / Sonnet 5
Sonnet 5 / Sonnet 5
0.4
0.6
0.8
1.0
(c) Sonnet 5 monitor
person / older
Sonnet 5 / older
person / Sonnet 5
Sonnet 5 / Sonnet 5
0.4
0.6
0.8
1.0
(d) Checked, caught at 2% FPR
person / older
Sonnet 5 / older
person / Sonnet 5
Sonnet 5 / Sonnet 5
0%
50%
100%